

GOVERNMENT OF KARNATAKA
PUBLIC WORKS DEPARTMENT (PWD)
AI-POWERED DPR COMPLIANCE, RISK ASSESSMENT & APPROVAL MANAGEMENT SYSTEM

OFFICIAL GOVERNMENT ORDER & TECHNICAL SANCTION DOSSIER

BANGALORE OUTER RING ROAD

Application Reference: KPWD-DPR-2026-717F86 · Government Sanction Order: KPWD/SANCTION/2026/GO-717F86

Project Sector	Infrastructure	Estimated Cost	■ 500.00 Crores
Location / District	Bengaluru Urban, Bengaluru Urban	Implementation Period	16 Months
Submitting Authority	Test User	Submission Date	2026-09-02
Sanctioning Authority	Principal Secretary, PWD Karnataka	Sanction Date	2026-09-03
Overall Status	FINAL APPROVED & SANCTIONED	Digital Verification ID	51B0055ACB86552E



GOVERNMENT DIGITAL VERIFICATION SEAL & AUDIT HASH

This Detailed Project Report has completed full 5-Directorate Technical, Financial, Compliance, Risk, and Executive appraisal under Karnataka PWD Code. All itemized rates, structural pavements, and environmental clearances are verified and digitally signed.

Digital SHA-256 Hash: 51B0055ACB86552E

Verification Portal: <https://kpwd.karnataka.gov.in/verify?hash=51B0055ACB86552E>

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1.0 EXECUTIVE SUMMARY & SANCTION DECISION

Executive Briefing for Bangalore outer ring road

- **Project Scope & Rationale**:** Comprehensive infrastructure development project under ****Karnataka Public Works Department (PWD)**** with a total capital investment outlay of ****₹500.00 Crores****. The project serves as a crucial economic freight and passenger link in Bengaluru Urban, designed to upgrade existing sub-standard pavement geometry, eliminate bottlenecks, and enhance inter-district transit reliability.
- **Technical Viability**:** Engineered in accordance with ****IRC:37-2018**** flexible pavement guidelines for a design life of 15 years with an effective subgrade CBR of ****8.0%**** and ****45 MSA**** traffic intensity.
- **Financial Benchmarking**:** Benchmarked against ****Karnataka PWD Schedule of Rates (KPWD SR 2025-26)**** with civil construction works accounting for ****₹360.0 Cr (72%)**** and direct land purchase under LARR 2013 at ****₹70.0 Cr****.
- **Strategic Recommendation**:** Recommended for ****Administrative Approval & Technical Sanction**** subject to fulfillment of Stage-I Forest clearance and utility shifting milestones.

2.0 DPR SUBMISSION & APPLICANT PARTICULARS

Applicant / Submitter Name	Test User	Employee / User ID	KPWD-ENG-4892
Submitting Department	PWD Highways Division, Karnataka	Designation	Executive Engineer (Roads)
Project Category	Infrastructure	District & Taluk	Bengaluru Urban / Central
Expected Start Date	2026-10-01	Target Completion Date	2028-09-30

3.0 DEPARTMENT-WISE COMPREHENSIVE REVIEW SUMMARY

Directorate	Assigned Reviewing Officer	Status	Approval Date	Officer Remarks & Certification
Technical Department Review	Director of Technical Directorate Chief Engineer	APPROVED	2026-09-03 02:57	Certified compliant with Karnataka PWD Standards.
Financial Department Review	Director of Financial Directorate Financial Controller	APPROVED	2026-09-03 02:57	Certified compliant with Karnataka PWD Standards.
Compliance Department Review	Director of Compliance Directorate Compliance Director	APPROVED	2026-09-03 02:57	Certified compliant with Karnataka PWD Standards.
Risk Assessment Review	Director of Risk Directorate Director (Quality Audit)	APPROVED	2026-09-03 02:57	Certified compliant with Karnataka PWD Standards.
Executive Authority Sanction	Director of Executive Directorate Principal Secretary	APPROVED	2026-09-03 02:57	Certified compliant with Karnataka PWD Standards.

4.0 SEQUENTIAL MULTI-LEVEL APPROVAL WORKFLOW TIMELINE

Stage 1: Technical Review ✓ Chief Engineer Approved CBR 8.5% · IRC:37-2018 Crust	Stage 2: Financial Review ✓ Chief Accounts Officer Approved KPWD SoR 2025-26 BOQ Audit	Stage 3: Compliance Review ✓ Legal Director Approved FCA 1980 · SEIAA Category-B
Stage 4: Quality & Risk Review ✓ Director QA Approved IRC:SP:13 HFL Drainage · Safety	Stage 5: Executive Sanction ✓ Principal Secretary Approved Official GO Sanction Issued	Final Approval Conferred ■ Technical Sanction Granted 100% 5-Directorate Certified

5.0 AI DOCUMENT QUALITY & COMPLETENESS INDEX (DQCI)

Engineering Assessment Dimension	Score	Max	Audit Feedback & Observation
Executive Summary & Objectives	12	15	Project background, objectives, and scope clearly articulated.
Geotechnical & Soil Test Reports	10	15	Subgrade CBR values and geotechnical parameters verified against IRC standards.
Traffic Survey & Axle Load Data	15	15	Traffic intensity and projected Million Standard Axles (MSA) provided.
Pavement & Structural Design	15	15	Pavement layer crust thicknesses and structural cross-sections detailed.
BOQ Cost Estimate & Schedule of Rates	12	15	Itemized cost breakdown benchmarked to Karnataka PWD Schedule of Rates.
Statutory & Environmental Clearances	10	10	Statutory clearances checklist and environmental management plan outlined.
Engineering Drawings & Visual Assets	10	10	21 visual diagrams, maps, and technical drawings attached.
Document Formatting & Integrity	5	5	Structured layout across 47 pages with 11,505 extracted words.

6.0 COMPREHENSIVE RISK ASSESSMENT & MITIGATION ROADMAP

Risk Category	Description	Impact	Probability	Mitigation Strategy
Right-of-Way (RoW) Encroachment	Linear ribbon development along highway stretches	Medium	Low	Direct consent purchase under LARR 2013 with boundary stone fixing
Monsoon Drainage Flooding	High intensity storm precipitation causing waterlogging	Medium	Medium	Trapezoidal side drains designed for 50-year return period per IRC:SP:13
Utility Relocation Shifting	KPTCL 66kV transmission towers and water supply mains	Medium	Medium	Deposit work payments credited upfront to utility departments with joint tracking
Bitumen Star-Rate Price Rise	Global crude oil price fluctuations affecting VG-30 bitumen	Low	Low	KPWD standard Star-Rate variation clause linked to IOCL refinery price indices

7.0 ACTIONABLE AI RECOMMENDATIONS & COST OPTIMIZATIONS

Recommendation Title	Expected Benefit & Optimization
Utilize Reclaimed Asphalt Pavement (RAP) in WMM/DBM	Save ■1.85 Cr in raw aggregate procurement while satisfying IRC:120-2015 green highway standards.
Precast Concrete Box Culverts for Cross-Drainage	Reduces monsoon bridge construction downtime by 45 calendar days.
Install Solar LED Delineators & High-Intensity Retroreflective Signboards	Conforms to IRC:67-2022 safety guidelines, reducing night-time accident probability.

8.0 FINANCIAL ANALYSIS & SCHEDULE OF RATES (SoR) AUDIT

Cost Component	Allocated Outlay	Share (%)	Benchmark Schedule
Civil Construction Works (BOQ Items)	■ 360.00 Cr	72.0%	KPWD Schedule of Rates 2025-26
Land Acquisition & Resettlement (LARR 2013)	■ 80.00 Cr	16.0%	State Multiplier & Solatium
Utility Relocation (KPTCL / Water Mains)	■ 35.00 Cr	7.0%	Joint Utility Estimate
Contingencies, Quality Control & QC Audit (5%)	■ 25.00 Cr	5.0%	PWD Code Provisions
TOTAL SANCTIONED CAPITAL BUDGET	■ 500.00 Cr	100.0%	Administrative Sanction GO

9.0 TECHNICAL PAVEMENT, GEOTECHNICAL & STRUCTURAL ASSESSMENT

Technical & Geotechnical Specifications:

- 1. ****Carriageway & Geometric Design****: Proposed 7.0m Carriageway + 1.5m Paved Shoulders with a design speed of 100 km/h conforming to IRC:73/IRC:86 geometric standards.
- 2. ****Pavement Composition****: Total design crust comprising Wearing Course (BC): 40 mm, Binder Course (DBM): 100 mm, Base Course (WMM): 250 mm, Sub-Base (GSB): 200 mm.
- 3. ****Soil & Subgrade Properties****: Laboratory CBR testing satisfies NABL guidelines with effective design subgrade CBR of 8.0%.
- 4. ****Cross-Drainage & Hydraulic Structures****: Inventory includes ****12 Major Bridges****, ****13 Minor Bridges****, and ****24 Precast Box Culverts**** designed for 100-year return period High Flood Level (HFL) per IRC:SP:13.
- 5. ****Road Safety & ITS****: Edge crash barriers (W-beam), solar LED delineators, retro-reflective signage per IRC:67, and junction channelization.

10.0 STATUTORY, FOREST & ENVIRONMENTAL CLEARANCES MATRIX

Statutory Clearance Category	Governing Act	Compliance Status	Action / Condition
Forest Land Diversion	Forest Conservation Act (FCA 1980)	Stage-I In-Principle Approved	Compensatory Afforestation land transferred
Environmental Clearance	EIA Notification 2006 (SEIAA)	Category B Appraisal Cleared	Environmental Management Plan (EMP) budgeted
Tree Felling Scheme	Karnataka Tree Preservation Act	Permission Granted	1:10 Compensatory Plantation Scheme
Waterbody Crossing	Water Resources Department	Hydraulic NOC Issued	High Flood Level (HFL) clearances verified

11.0 PROJECT EXECUTION READINESS & AI CONFIDENCE

Technical Readiness 95% Design Vetted	Financial Readiness 90% Budget Allotted	Compliance Readiness 98% NOCs Obtained	Execution Readiness 92% Tender Ready
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12.0 OFFICIAL GOVERNMENT SANCTION ORDER & DIGITAL CERTIFICATION

GOVERNMENT OF KARNATAKA — PUBLIC WORKS DEPARTMENT ORDER

Sanction Order No: KPWD/SANCTION/2026/GO-717F86 | Date: 03-Sep-2026

Administrative Approval and Technical Sanction are hereby accorded under Karnataka Public Works Department Code for the execution of Detailed Project Report titled **Bangalore outer ring road** at an estimated outlay of **₹ 500.00 Crores**. The Chief Engineer (Highways) is authorized to invite tenders and commence work in adherence to KPWD quality assurance protocols.

Scrutinized & Verified By: Chief Engineer (Technical) Er. R. S. Patil, FIE [DIGITALLY SIGNED]	Financial Concurrence: Chief Accounts Officer Shri M. Venkatesh, KAS [DIGITALLY SIGNED]	Approved & Sanctioned: Principal Secretary (PWD) Dr. S. Radhakrishnan, IAS [DIGITALLY SIGNED]
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